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DADS7275: Milestone 1 - Proposal
Due: 2/8/2026

Wind Turbine Failure Prediction Using SCADA Data

1 Project Overview and Objectives

Wind turbines are critical assets in modern renewable energy systems, yet unexpected component failures can result in costly downtime, expensive repairs, and lost power generation. Traditional maintenance strategies are often reactive or schedule-based and do not fully leverage the rich sensor data collected by turbine supervisory control and data acquisition (SCADA) systems. As wind energy deployment continues to expand, there is a growing need for reliable, data-driven early-warning systems capable of detecting abnormal operating conditions before failures occur.

The objective of this project is to develop and evaluate supervised machine learning classification models that predict whether a wind turbine is likely to experience a fault within a short future time window (e.g., 24–48 hours) using historical SCADA telemetry data. The primary goals of the project are to:

- Formulate turbine fault prediction as a binary classification problem (failure vs. normal operation).
- Engineer temporal features that capture early degradation patterns in multivariate sensor data.
- Compare multiple classification algorithms with respect to predictive performance and interpretability.
- Evaluate model performance using metrics aligned with engineering decision-making, with particular emphasis on detecting failure events.

2 Dataset Description and Suitability

Data Source

The dataset used in this project is publicly available and accessible without special permissions.

> Kasimov, A. (2024). *Wind turbine SCADA data for early fault detection* [Data set]. Zenodo. https://zenodo.org/records/10958775

The dataset is also mirrored on Kaggle, ensuring reproducibility and ease of access.

Dataset Structure and Features

The dataset consists of multivariate, time-stamped SCADA measurements collected at regular intervals from multiple wind turbines operating across several wind farms. Each observation represents a snapshot of turbine operating conditions at a given time and includes an associated operational or fault status label.

Key feature categories include:

- **Environmental variables:** wind speed, ambient temperature
- **Mechanical variables:** rotor speed, generator speed
- **Electrical variables:** power output, voltage, current
- **Thermal variables:** gearbox and bearing temperatures
- **Operational indicators:** anomaly or fault labels

The dataset contains thousands of observations across turbines, providing sufficient temporal depth and feature richness to support supervised classification and feature engineering.

Dataset Suitability and Challenges

This dataset is well-suited for the proposed analysis due to its multivariate time-series structure and labeled fault events. Anticipated challenges include class imbalance (failure events are relatively rare), missing or noisy sensor readings, and uncertainty in defining the optimal prediction horizon for near-term failures. These challenges will be addressed through appropriate preprocessing, feature engineering, and evaluation strategies.

3 Methodology

Preprocessing and Exploratory Analysis

Data preprocessing will include handling missing values, capping or removing extreme outliers, standardizing numerical features where appropriate, and ensuring correct chronological ordering of observations. Exploratory data analysis (EDA) will be conducted to examine sensor behavior during normal operation versus pre-failure periods, assess feature distributions, and identify correlations among variables.

Feature Engineering

To capture temporal degradation patterns, rolling statistical features such as moving means, standard deviations, and rates of change will be generated over multiple time windows. Lagged variables will be included to reflect recent operating history. Feature selection techniques, including correlation analysis and model-based importance measures, will be used to reduce redundancy. Dimensionality reduction methods (e.g., principal component analysis) may be explored if the feature space becomes excessively large.

Supervised Learning Models

The project is explicitly framed as a supervised binary classification task, aligning with core course content on foundational classification algorithms. The following models will be implemented and compared:

- **Logistic Regression** as an interpretable baseline classifier
- **Random Forest** to capture nonlinear relationships and feature interactions
- **Gradient Boosting** to evaluate high-performance ensemble classification

A time-aware train/validation/test split will be used to prevent information leakage from future observations. Class imbalance will be addressed using techniques such as class weighting or resampling where appropriate.

Evaluation Metrics

Model performance will be evaluated using accuracy, precision, recall, F1-score, and area under the ROC curve (AUC). Emphasis will be placed on recall for the failure class, as missed failures have high operational and financial costs in real-world turbine maintenance scenarios.

4 Expected Outcomes

This project is expected to produce a validated machine learning pipeline capable of estimating near-term wind turbine failure risk from SCADA data. The analysis should identify which sensor variables provide the strongest early-warning signals and demonstrate how classification-based data mining techniques can support condition-based maintenance strategies in renewable energy systems. Anticipated challenges such as class imbalance and sensor noise will be addressed through careful feature engineering, model selection, and evaluation design. Overall, the project will illustrate the practical value of supervised classification methods for improving the reliability and cost-effectiveness of renewable energy infrastructure.

5 AI Use Disclosure

AI tools were used to assist with refining proposal structure, organization, and language clarity. All technical decisions, dataset selection, modeling approaches, and methodological justifications reflect the author's own understanding and judgment. The final submission has been reviewed to ensure accuracy and compliance with Northeastern University's Academic Integrity and Use of AI policy.